MILESTONE AND TIMELINES

FYP I — PLANNED VS DELIVERED

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Status update against the FYP1 Gantt chart presented at Proposal Defence

No	Week(s)	Task / Milestone	Deliverable	Status
1	1–2	**Topic Confirmation & Concept Document v1**	Concept Document v1	✓ Completed
2	3–4	**Literature Review — 30 papers screened, 20 selected**	Literature Review Matrix (20 papers, 7 domains)	✓ Completed
3	5–6	**Shortlist 3 candidate models per engine; source datasets**	Model shortlist, datasets sourced	✓ Completed
4	7	**ML Model Selection & Concept Document v2**	Concept Document v2	✓ Completed
5	8	**Proposal Defence Preparation**	Proposal Defence slides	✓ Completed
6	9	**Proposal Defence**	Proposal Defence — submitted	✓ Completed
7	10–12	**Interim Report Writing**	Interim Report — submitted	✓ Completed

AHEAD OF SCHEDULE

Engine 1 (ARIMA / Prophet / LSTM) and Engine 2 (Random Forest / XGBoost / LightGBM) — training, evaluation, and SHAP explainability — plus an initial dashboard mockup, were completed during FYP1, once the Kaggle dataset and synthetic BNPL data were ready. This work was originally scheduled for FYP2 Weeks 1–6.

Winners selected: ARIMA (Engine 1, lowest RMSE) · XGBoost (Engine 2, highest F1 + ROC-AUC)

FYP II TIMELINE — TECHNICAL & DELIVERY

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Reconciled with the official UTP FYP II submission schedule

No	Activities	1 Sep 7–13	2 Sep 14–20	3 Sep 21–27	4 Sep 28– Oct 4	5 Oct 5–11	6 Oct 12–18	7 Oct 19–25	8 Oct 26– Nov 1	9 Nov 2–8	10 Nov 9–15	11 Nov 16–22	12 Nov 23–27	13 Nov 30– Dec 6	14 Dec 7–13	15 Dec 14–20
1	ARIMA Model Deployment & API Integration															
2	XGBoost + SHAP Deployment & API															
3	FastAPI Backend Development															
4	Next.js Dashboard – 4 Screens															
5	Full System Integration															
6	Technical/System Testing (self-QA)															
7	Cloud Deployment (Vercel + Render)															
8	UAT Preparation (demo profiles, recruit testers)															
9	UAT Sessions (5–7 real users)															
10	UAT Analysis (SUS + qualitative synthesis)															
11	Dissertation & Technical Paper Writing															
12	Presentation Prep & Final Demo															

Note: Week 8 is buffer time, and Week 9 is fully dedicated to writing the draft dissertation with results already in hand.

FYP II TIMELINE — TECHNICAL & DELIVERY

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No	Activities	Week	Description
1	**ARIMA Model Deployment & API Integration**	1-2	wrap the already-trained model in a FastAPI endpoint (/forecast) that takes user data, returns predicted future cash flow.
2	**XGBoost + SHAP Deployment & API**	1-2	same idea for risk: an endpoint (/risk-score) returning a risk label plus SHAP values explaining which features drove it.
3	**FastAPI Backend Development**	2-4	(everything that stores, processes, computes) — everything beyond the two ML endpoints: CRUD for transactions/BNPL plans, Supabase connection, CSV/statement parsing.
4	**Next.js Dashboard – 4 Screens**	3-5	(what the user sees and clicks) — Dashboard, Transactions, BNPL Manager, Risk Checker.
5	**Full System Integration**	5-6	connect frontend to real backend endpoints (replace placeholder data with live calls).
6	**Technical/System Testing (self-QA)**	6	unit tests, model sanity checks, deploy live (Vercel = frontend, Render = backend), then UAT.
7	**Cloud Deployment (Vercel + Render)**	6	
8	**UAT Preparation (demo profiles, recruit testers)**	5-6	
9	**UAT Sessions (5–7 real users)**	7	
10	**UAT Analysis (SUS + qualitative synthesis)**	7	
11	**Dissertation & Technical Paper Writing**	4-14	full FYP2 dissertation + separate short IEEE-format technical paper.
12	**Presentation Prep & Final Demo**	10-12	final slides + rehearsed live demo.

MILESTONES — GANTT VIEW

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Same 15-week grid as the technical timeline, for direct side-by-side comparison

No	Milestone	1 Sep 7–13	2 Sep 14–20	3 Sep 21–27	4 Sep 28–Oct 4	5 Oct 5–11	6 Oct 12–18	7 Oct 19–25	8 Oct 26–Nov 1	9 Nov 2–8	10 Nov 9–15	11 Nov 16–22	12 Nov 23–27	13 Nov 30–Dec 6	14 Dec 7–13	15 Dec 14–20
1	**Submission of Progress Assessment 1** *~5 weeks away*															
2	**Submission of Draft Dissertation and Draft Technical Paper** *~8 weeks away*															
3	**Submission of Dissertation (Soft Bound) and Revised Technical Paper** *~9 weeks away*															
4	**Project Presentation** *~11 weeks away*															
5	**Submission of Progress Assessment 2** *~11 weeks away*															
6	**Submission of Project Dissertation (Hard Bound)** *~13 weeks away*															

Fixed university deadlines

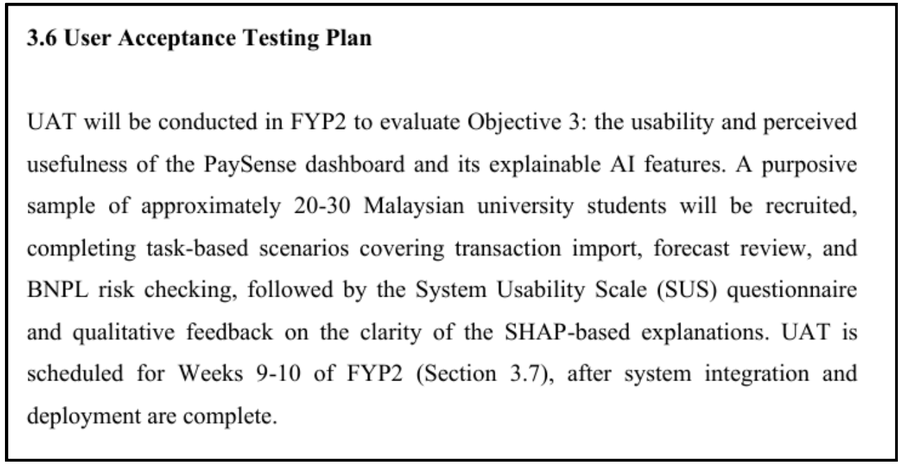
USER ACCEPTANCE TESTING (UAT)

UAT SAMPLE SIZE

Reducing UAT sample size — 20–30 → 5–7 targeted testers

Proposal: Transition from an untargeted pool of 20–30 users to 5–7 purposive testers grouped into 3 distinct financial personas. This high-contrast approach exposes boundary conditions and edge cases far more effectively than testing a large, homogeneous group.

Group	Profile	What It Tests
1	Frequent BNPL Users	High-frequency debt warnings & risk sensitivity
2	Budget-Conscious / Fixed-Income Users	Baseline stability & low-risk classification
3	Irregular / Gig-Income Users	Forecast handling of income variance



3.6 User Acceptance Testing Plan

UAT will be conducted in FYP2 to evaluate Objective 3: the usability and perceived usefulness of the PaySense dashboard and its explainable AI features. A purposive sample of approximately 20-30 Malaysian university students will be recruited, completing task-based scenarios covering transaction import, forecast review, and BNPL risk checking, followed by the System Usability Scale (SUS) questionnaire and qualitative feedback on the clarity of the SHAP-based explanations. UAT is scheduled for Weeks 9-10 of FYP2 (Section 3.7), after system integration and deployment are complete.

UAT Plan from Interim Report

UAT SCOPE & USER FLOW

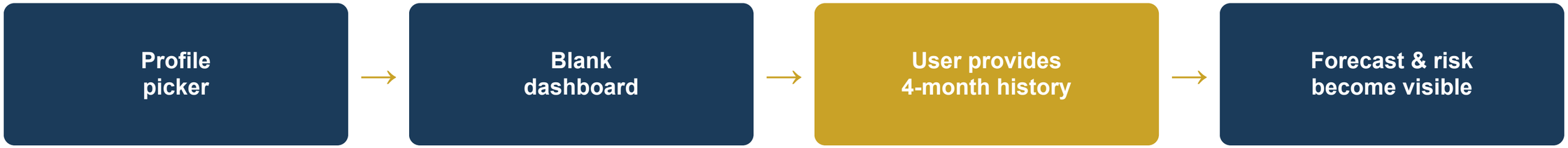
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What gets checked, and how a tester will actually move through the system

What UAT Checks (3 item)

i. Task-based scenario: transaction import
ii. Task-based scenario: forecast review
iii. Task-based scenario: BNPL risk checking
iv. System Usability Scale (SUS) questionnaire
v. Qualitative feedback on SHAP explanation clarity

User Flow



Demo data: 2–3 persona-matched profiles (not one generic dataset) — fixes the "everyone sees the same forecast" problem while avoiding real financial data. See Q4, previous slides.

Demo data approach for UAT

Since testers can't hand over real bank statements, and one generic demo dataset would mean everyone sees an identical forecast, I'll prepare 2–3 persona-matched demo profiles instead (matching the groups on the previous slide). Acceptable approach?
CSV / PDF data feeding : Most banks students use (Maybank, CIMB) issue statements as PDF, not CSV. Plan: accept both formats, with PDF converted to CSV via a bank-statement-conversion tool where direct parsing needs support.

OPTIONS — PROVIDING 4-MONTH HISTORY

How should a UAT tester get enough history for the forecast/risk features to work?

Option	What It Is	Trade-off	Verdict
1. One Generic Fake History	**Single pre-built** demo dataset, used identically for every tester	Every tester sees the exact same forecast graph — can't tell if the ML feels accurate to them	**Rejected**
2. Persona-Matched Histories (2–3 variants)	**2–3 distinct fake histories**, one **per persona** (heavy-BNPL, budget-conscious, gig-worker); each tester gets the one matching their group	Confirms the system behaves correctly for that situation — not that it matched their real life	**Recommended baseline**
3. Real Data, Stripped & Safeguarded	Testers upload a **real statement**; only amount/date/category kept, **raw file discarded immediately**, disclosed upfront	Still asks for real financial data (consent question); still carries live PDF-parsing risk	Only if UTP requires formal consent/ethics process anyway
4. Personalized Synthetic Data from Questionnaire	4–5 **quick questions** → generate a **detailed history** using the same logic as Engine 2's training data — grounded in their real numbers, **no real bank data** touched	More building work than Option 2	Stretch goal — if time allows

QUESTIONS FOR DISCUSSION

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Project scope — decisions needing sign-off before development continues

Q1. Can already-submitted in FYP1 content be revised in FYP2?

Updated as of now :

- Timelines Gantt chart
- Feature scope - User Authentication
- UAT Sample Size

Answer

Can.

Q2. User authentication scope

The Interim Report scoped “user authentication and multi-user account management” as **out of scope**. But real **multi-user UAT (5–7 testers)** needs some way to keep each tester's data separate. Acceptable to add a l**ightweight profile picker (no password) or a fixed username/password per tester** — without building full production authentication?

Answer

Put username in database. So, testing user will just enter username credential to log in.

SYNTHETIC DATASET — JUSTIFICATION

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Answering the supervisor's earlier review comment

Question raised previously

“Why was a real BNPL dataset not used? Why was synthetic data necessary? What limitations does synthetic data introduce? How representative is the generated dataset of actual Malaysian BNPL users?”

Why not real data

No Malaysian BNPL provider or credit bureau publishes individual-level default records — commercially confidential, so it simply doesn't exist to use.

Why synthetic is grounded, not arbitrary

Age distributions from Tan et al. (2026, N=150); employment/income brackets from Osman et al. (2024, N=440) — real Malaysian survey data, not invented numbers.

How it was built

Age & employment sampled from those real distributions; income/expenses sampled conditionally on employment (not independently random); BNPL behaviour built with a deliberate positive relationship to risk — mirrors real relationships, not random noise.

ADDITIONAL KNOWLEDGE NOTES

Concept — what the system actually does

- **Core purpose:** build an accurate enough picture of income/expenses to forecast future balance and judge whether a new BNPL purchase is safe.
- **Bank integration:** out of scope.
- **Missed BNPL payments:** currently, if due date passes without the user manually logging payment → defaults to Overdue. (The "auto cross-check next month's statement against pending dues" idea is undocumented — flag as open, not assumed.)
- **New user data feeding:** needs at least 4 months of past transaction history before forecast/risk features work.
- **Continuous data feeding:** monthly statement upload + manual entry (for untracked cash spending only).
- **Data feeding method:** (1) CSV, or (2) PDF statement — PDF can be converted to CSV via a tool like BankStatementConverter as a workaround.
- **UAT Analysis finishing =** "I now have real evidence for Objective 3, time to write it up" — not "the project is done."

QUALITY TEST

Full Testing checklist
(you do this yourself, before any real user touches it)

- **System functionality —** all 4 screens load & navigate correctly; manual transaction entry saves correctly; BNPL Plan creation (manual + auto from Risk Checker) calculates instalment schedules correctly.
- **Data import —** CSV import parses/maps correctly; PDF import tested against real Maybank/CIMB/RHB sample formats; preview/confirmation screen shows parsed data before saving; 4-month history boundary is correct (upload Oct 15 → captures back to June 15); decide + test how refunds/internal transfers are represented.
- **Model behavior (designed inputs, not live guessing) —** high savings + small purchase → Low Risk; forecast goes negative + large purchase → High Risk; SHAP explanations spot-checked against known cases (e.g. shap_waterfall_highrisk.png).
- **Integration & deployment —** dashboard pulls real backend data, not placeholders; live on Vercel+Render, tested as a stranger would on a fresh session.
- Bonus, self-check only — try it once on your own phone browser.

Full UAT checklist (real users — your friends)

- **Core, already scoped:** task completion (import statement → check forecast → run risk check, unguided); SUS questionnaire; qualitative feedback on explanation clarity.
- **Cheap free additions:** "Does this forecast feel realistic to you?"; "Would this warning have changed your decision?"
- **Only if scope is broadened + time allows:** live risk-tier scenario testing with real purchase amounts; recurring bill recognition (only if that feature actually gets built).
- **Not tested anywhere — already done:** model accuracy (RMSE/MAE/MAPE), validated in FYP1 against historical hold-out data.

THE END. THANK YOU.